

# Lecture 11

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## Necessary conditions for optimality

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## How do we know that a point is a local minimizer?

Let's review the different ways we can tell if a point is a local minimizer. We'll start with the univariate case.

# Univariate Minimizer Conditions

| Function Type | Local Minimizer                                                              | Global Minimizer                                                 | Reference |
|---------------|------------------------------------------------------------------------------|------------------------------------------------------------------|-----------|
| $C^0$         | $f(x) \geq f(x_*)$ in some neighborhood                                      | $f(x) \geq f(x_*)$ for all $x$                                   | Lec 2     |
| $C^1$         | $f'(x) < 0$ for $x < x_*$ and $f'(x) > 0$ for $x > x_*$ in some neighborhood | Compare all local minimizers                                     | Lec 2     |
| Convex $C^1$  | $f'(x_*) = 0$                                                                | Any local minimizer is global                                    | Thm 11.23 |
| $C^2$         | $f'(x_*) = 0$ and $f''(x_*) \geq 0$                                          | $f'(x_*) = 0$ , $f''(x_*) > 0$ , and $f''(x) \geq 0$ for all $x$ | Lec 2     |

# Multivariate Optimization Conditions

| Function Type | Local Minimizer                            | Global Minimizer               | Reference          |
|---------------|--------------------------------------------|--------------------------------|--------------------|
| $C^0$         | $f(x) \geq f(x_*)$ in<br>some neighborhood | $f(x) \geq f(x_*)$ for all $x$ | Lec 2              |
| $C^1$         | ??                                         | ??                             | Today's<br>lecture |
| Convex $C^1$  | ??                                         | ??                             | Today's<br>lecture |
| $C^2$         | ??                                         | ??                             | Today's<br>lecture |

Simply put, if a point is a local minimizer of a differentiable function, then the gradient must be zero.

### **Theorem (First Order Necessary Condition)**

*If  $\mathbf{x}_* \in \Omega$  is a local minimizer and  $f \in C^1(\Omega)$  then  $\nabla f(\mathbf{x}_*) = \mathbf{0}$ .*

Simply put, if a point is a local minimizer of a differentiable function, then the gradient must be zero.

### **Theorem (First Order Necessary Condition)**

*If  $\mathbf{x}_* \in \Omega$  is a local minimizer and  $f \in C^1(\Omega)$  then  $\nabla f(\mathbf{x}_*) = \mathbf{0}$ .*

To prove this, we will use a contradiction. Suppose that the gradient is not zero. Then we can find a direction in which the function decreases.

## Proof of the first order necessary condition

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Since  $\nabla f(\mathbf{x}_*) \neq \mathbf{0}$ , the norm is also nonzero (and positive):  $\|\nabla f(\mathbf{x}_*)\| > 0$ .

Next, we will use this to show that there is a neighborhood of  $\mathbf{x}_*$  in which the function decreases.

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Since  $\nabla f$  is continuous at  $\mathbf{x}_*$ , there exists a neighborhood around  $\mathbf{x}_*$  in which the  $\nabla f(\mathbf{x}_*)^\top \mathbf{p}$  is also negative.

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That is, we can choose  $\delta > 0$  such that for all  $t \in [0, \delta]$ ,

$$\nabla f(\mathbf{x}_* + t\mathbf{p})^\top \mathbf{p} < 0$$

Then we can use Taylor's theorem to calculate the function value at  $\mathbf{x}_* + t\mathbf{p}$ .

Remember, Taylor's theorem is the multivariate version of the mean value theorem. It tells us that there is a point between  $\mathbf{x}_*$  and  $\mathbf{x}_* + t\mathbf{p}$  with exactly the right gradient to exactly write the function value at  $\mathbf{x}_* + t\mathbf{p}$ :

$$f(\mathbf{x}_* + t\mathbf{p}) = f(\mathbf{x}_*) + t\nabla f(\mathbf{x}_* + \xi\mathbf{p})^\top \mathbf{p}$$

for some  $\xi \in (0, t)$ .



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for some  $\xi \in (0, t)$ .

Then because we have chosen  $\delta$  to be small enough that the gradient is bounded away from zero, we have

$$f(\mathbf{x}_* + t\mathbf{p}) < f(\mathbf{x}_*)$$

for all  $t \in (0, \delta]$ .

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Then because we have chosen  $\delta$  to be small enough that the gradient is bounded away from zero, we have

$$f(\mathbf{x}_* + t\mathbf{p}) < f(\mathbf{x}_*)$$

for all  $t \in (0, \delta]$ .

But then  $\mathbf{x}_*$  is not a local minimizer, a contradiction!

# Multivariate Optimization Conditions

| Function Type | Local Minimizer                                       | Global Minimizer               | Reference       |
|---------------|-------------------------------------------------------|--------------------------------|-----------------|
| $C^0$         | $f(x) \geq f(x_*)$ in some neighborhood               | $f(x) \geq f(x_*)$ for all $x$ | Lec 2           |
| $C^1$         | $\nabla f(x_*) = 0$<br>(necessary but not sufficient) | Compare all local minimizers   | Theorem 1.1     |
| Convex $C^1$  | ??                                                    | ??                             | Today's lecture |
| $C^2$         | ??                                                    | ??                             | Today's lecture |

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### Definition (Descent direction)

Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be differentiable at  $\mathbf{x}$ . A vector  $\mathbf{p}$  is a **descent direction** of  $f$  at  $\mathbf{x}$  if there exists  $\delta > 0$  such that  $f(\mathbf{x} + t\mathbf{p}) < f(\mathbf{x})$  for all  $t \in (0, \delta)$ .

If  $\mathbf{x}$  is not a stationary point, then we can always find a descent direction.

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$$\mathbf{p} = -\nabla f(\mathbf{x}).$$

This is guaranteed not to be zero, since  $\mathbf{x}$  is not a stationary point.

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(We can't have a maximizer or a saddle point in a function that “curves up”)

And we will show below that for convex functions, any local minimizer is a global minimizer.

So, if we can find a stationary point, we know it's a global minimizer.

Formally,

### **Theorem**

*Let  $f : \Omega \rightarrow \mathbb{R}$  be convex. Then every local minimizer is a global minimizer.*

*Furthermore, if  $f \in C^1(\Omega)$ , then  $\nabla f(\mathbf{x}_*) = \mathbf{0}$  implies  $\mathbf{x}_*$  is a global minimizer, i.e. a stationary point is automatically a global minimizer.*

| Function Type | Local Minimizer                                       | Global Minimizer               | Reference       |
|---------------|-------------------------------------------------------|--------------------------------|-----------------|
| $C^0$         | $f(x) \geq f(x_*)$ in some neighborhood               | $f(x) \geq f(x_*)$ for all $x$ | Lec 2           |
| $C^1$         | $\nabla f(x_*) = 0$<br>(necessary but not sufficient) | Compare all local minimizers   | Theorem 1.1     |
| Convex $C^1$  | $\nabla f(x_*) = 0$                                   | Any local minimizer is global  | Theorem 1.2     |
| $C^2$         | ??                                                    | ??                             | Today's lecture |



## Proof of the first part of the theorem

Here we will prove the first part of the theorem, that if  $f$  is convex then any local minimizer is a global minimizer.

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We will do this by contradiction. We will suppose that there exists a point  $\mathbf{x}_*$  that is a local minimizer but not a global minimizer.

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We will do this by contradiction. We will suppose that there exists a point  $\mathbf{x}_*$  that is a local minimizer but not a global minimizer.

We will show that this leads to a contradiction, that  $\mathbf{x}_*$  cannot in fact be a local minimizer.

Suppose  $\mathbf{x}_*$  is a local but not global minimizer.

Then there is some point  $\mathbf{z} \in \Omega$  such that  $f(\mathbf{z}) < f(\mathbf{x}_*)$ .

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We will consider the line segment connecting  $\mathbf{x}_*$  and  $\mathbf{z}$ . Because the function is convex, any point on this line segment must have a function value less than or equal to the straight line connecting the function values at the endpoints.

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We will consider the line segment connecting  $\mathbf{x}_*$  and  $\mathbf{z}$ . Because the function is convex, any point on this line segment must have a function value less than or equal to the straight line connecting the function values at the endpoints.

Since we've said that  $f(\mathbf{z}) < f(\mathbf{x}_*)$ , this means that any point on the line segment must also have a function value less than  $f(\mathbf{x}_*)$ .

Formally, for any  $t \in (0, 1)$ ,

$$\begin{aligned} f(t\mathbf{z} + (1-t)\mathbf{x}_*) &\leq tf(\mathbf{z}) + (1-t)f(\mathbf{x}_*) \\ &< tf(\mathbf{x}_*) + (1-t)f(\mathbf{x}_*) \\ &= f(\mathbf{x}_*) . \end{aligned}$$

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OK, so this means that as we move along the line segment connecting  $\mathbf{x}_*$  and  $\mathbf{z}$ , the function value will *decrease*.



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OK, so this means that as we move along the line segment connecting  $\mathbf{x}_*$  and  $\mathbf{z}$ , the function value will *decrease*.

Intuitively, we can see that this means the point cannot be a local minimizer, since the function decreases around it. But let's make this more formal.

Formally, for any neighborhood  $N$  of  $\mathbf{x}_*$ , we can pick a sufficiently small  $t$  such that moving along the line segment by an amount  $t$  will take us to a point  $\mathbf{x}(t)$  that is in the neighborhood  $N$ :

$$\mathbf{x}(t) := \mathbf{x}_* + t(\mathbf{z} - \mathbf{x}_*) \in N$$

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And since we've said that  $f(\mathbf{z}) < f(\mathbf{x}_*)$ , this means that  $f(\mathbf{x}(t)) < f(\mathbf{x}_*)$  for all  $t$  in this neighborhood.

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$$\mathbf{x}(t) := \mathbf{x}_* + t(\mathbf{z} - \mathbf{x}_*) \in N$$

And since we've said that  $f(\mathbf{z}) < f(\mathbf{x}_*)$ , this means that  $f(\mathbf{x}(t)) < f(\mathbf{x}_*)$  for all  $t$  in this neighborhood.

This contradicts the fact that  $\mathbf{x}_*$  is a local minimizer.

## Proof of the second part of the theorem

We have shown that if  $f$  is convex, then any local minimizer is a global minimizer.

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We have shown that if  $f$  is convex, then any local minimizer is a global minimizer.

Now we will show that if  $f$  is convex and differentiable, then a stationary point must be a global minimizer.

We will suppose that there exists a point  $\mathbf{x}_*$  that is a stationary point but not a global minimizer.

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Then we'll show that there must be a direction in which the function decreases, so the point cannot be a stationary point.



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Then we'll show that there must be a direction in which the function decreases, so the point cannot be a stationary point.

Specifically, we'll consider the point  $\mathbf{z}$  that *is* the global minimizer. We'll note that the line segment connecting  $\mathbf{x}_*$  and  $\mathbf{z}$  must have a negative slope.

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Specifically, we'll consider the point  $\mathbf{z}$  that *is* the global minimizer. We'll note that the line segment connecting  $\mathbf{x}_*$  and  $\mathbf{z}$  must have a negative slope.

Then, we'll argue that the function must travel below the line segment (since it's convex), so its slope must also be negative. (draw it)

Formally: suppose that  $f \in C^1(\Omega)$  and  $\nabla f(\mathbf{x}_*) = \mathbf{0}$ .

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If  $\mathbf{x}_*$  is not a global minimizer, then there is some point  $\mathbf{z} \in \Omega$  such that  $f(\mathbf{z}) < f(\mathbf{x}_*)$ .

Then we can find the directional derivative of  $f$  at  $\mathbf{x}_*$  in the direction of  $\mathbf{z} - \mathbf{x}_*$ :

$$\begin{aligned} & \left. \frac{d}{dt} f(\mathbf{x}_* + t(\mathbf{z} - \mathbf{x}_*)) \right|_{t=0} \\ &= \lim_{t \rightarrow 0^+} \frac{f(\mathbf{x}_* + t(\mathbf{z} - \mathbf{x}_*)) - f(\mathbf{x}_*)}{t} && \text{(definition)} \\ &= \lim_{t \rightarrow 0^+} \frac{f(t\mathbf{z} + (1-t)\mathbf{x}_*) - f(\mathbf{x}_*)}{t} && \text{(re-arranging)} \\ &\leq \lim_{t \rightarrow 0^+} \frac{tf(\mathbf{z}) + (1-t)f(\mathbf{x}_*) - f(\mathbf{x}_*)}{t} && \text{(convexity)} \\ &= \lim_{t \rightarrow 0^+} \frac{tf(\mathbf{z}) - tf(\mathbf{x}_*)}{t} && \text{(simplifying)} \\ &= f(\mathbf{z}) - f(\mathbf{x}_*) < 0 \end{aligned}$$

## Second order conditions

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## Second order necessary conditions

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If we have a  $C^2$  function, then we can use the second derivative to get a necessary condition for point to be a local minimizer.

It's very simple: the gradient must be zero, and the Hessian must be positive semidefinite.

Formally:

**Theorem (Second Order Necessary Conditions)**

*Let  $f \in C^2(\Omega)$ . If  $\mathbf{x}_* \in \Omega$  is a local minimizer, then*

$$\nabla f(\mathbf{x}_*) = \mathbf{0} \quad \text{and} \quad \nabla^2 f(\mathbf{x}_*) \succeq 0.$$





# Multivariate Optimization Conditions

| Function Type | Local Minimizer                                                                       | Global Minimizer               | Reference   |
|---------------|---------------------------------------------------------------------------------------|--------------------------------|-------------|
| $C^0$         | $f(x) \geq f(x_*)$ in some neighborhood                                               | $f(x) \geq f(x_*)$ for all $x$ | Lec 2       |
| $C^1$         | $\nabla f(x_*) = 0$<br>(necessary but not sufficient)                                 | Compare all local minimizers   | Theorem 1.1 |
| Convex        | $\nabla f(x_*) = 0$                                                                   | Any local minimizer is global  | Theorem 1.2 |
| $C^2$         | $\nabla f(x_*) = 0$ and $\nabla^2 f(x_*) \succeq 0$<br>(necessary but not sufficient) | ??                             | Theorem 2.1 |

## Proof of the second order necessary condition

We will do this proof similar to the proof of the first order necessary condition. We'll show that if the Hessian is not positive semidefinite, then we can find a direction in which the function decreases, so the point cannot be a local minimizer.

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We will do this proof similar to the proof of the first order necessary condition. We'll show that if the Hessian is not positive semidefinite, then we can find a direction in which the function decreases, so the point cannot be a local minimizer.

The first equality is true by the first order necessary condition.

Now suppose for contradiction that  $\nabla^2 f(\mathbf{x}_*)$  not positive semidefinite.

Then there is some nonzero  $\mathbf{p} \in \mathbb{R}^n$  such that  $\mathbf{p}^\top \nabla^2 f(\mathbf{x}_*) \mathbf{p} < 0$ .

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Then there is some nonzero  $\mathbf{p} \in \mathbb{R}^n$  such that  $\mathbf{p}^\top \nabla^2 f(\mathbf{x}_*) \mathbf{p} < 0$ .

Since  $\nabla^2 f$  is continuous at  $\mathbf{x}_*$ , there exists  $\delta > 0$  such that

$$\mathbf{p}^\top \nabla^2 f(\mathbf{x}_* + t\mathbf{p}) \mathbf{p} < 0$$

for all  $t \in [0, \delta]$ .

By Taylor's theorem, for all  $t \in (0, \delta)$ ,

$$f(\mathbf{x}_* + t\mathbf{p}) = f(\mathbf{x}_*) + \underbrace{t\nabla f(\mathbf{x}_*)^\top}_{=0} \mathbf{p} + \underbrace{\frac{t^2}{2} \mathbf{p}^\top \nabla^2 f(\mathbf{x}_* + \xi t\mathbf{p}) \mathbf{p}}_{<0} < f(\mathbf{x}_*)$$

for  $\xi \in (0, 1)$  and  $t$  nonzero. Hence  $\mathbf{x}_*$  is not a local minimizer.  $\therefore$

In the proofs of Theorems Theorem 1.1 and Theorem 2.1, we have used the fact that if a univariate continuous function is positive (resp. negative) at a point  $t_0 \in \mathbb{R}$ , then it must be positive (resp. negative) in a neighborhood of  $t_0$ .

This is almost trivial: say  $g(t_0) > 0$ , then pick  $\varepsilon = g(t_0)/2$ , by continuity at  $t_0$  there exists  $\delta > 0$  so that whenever  $|t - t_0| < \delta$ , we have  $|g(t) - g(t_0)| < g(t_0)/2$

i.e.,  $-g(t_0)/2 < g(t) - g(t_0) < g(t_0)/2$ .

So  $g(t) > g(t_0)/2 > 0$  for all  $t \in (t_0 - \delta, t_0 + \delta)$ .

In the proof of Theorem 1.1, we applied this to the continuous univariate function of  $t$

$$g(t) = \nabla f(\mathbf{x}_* + t\mathbf{p})^\top \mathbf{p}$$

and in the proof of Theorem 2.1, we applied it to

$$g(t) = \mathbf{p}^\top \nabla^2 f(\mathbf{x}_* + t\mathbf{p}) \mathbf{p}$$



What if we have matrix-variate functions in Theorems Theorem 1.1 and Theorem 2.1?

What if we have matrix-variate functions in Theorems Theorem 1.1 and Theorem 2.1?

Recall that the notion of positive semidefiniteness applies to all linear maps on an inner product space. The Hessian  $\nabla^2 f(X) : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$  is positive semidefinite if

$$\text{tr}(H^\top [\nabla^2 f(X)](H)) \geq 0 \quad \text{for all } H \in \mathbb{R}^{m \times n};$$

and positive definite if

$$\text{tr}(H^\top [\nabla^2 f(X)](H)) > 0 \quad \text{for all } H \in \mathbb{R}^{m \times n}.$$

Likewise for  $\mathbb{S}^n$  in place of  $\mathbb{R}^{m \times n}$ .

## Sufficient conditions for optimality

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### Theorem (First Order Sufficient Conditions)

*Let  $f$  be continuous at  $\mathbf{x}_* \in \mathbb{R}^n$  and differentiable in a neighborhood of  $\mathbf{x}_*$ . If there is a  $\delta > 0$  with*

$$\nabla f(\mathbf{x}_* + \mathbf{p})^\top \mathbf{p} > 0$$

*whenever  $\|\mathbf{p}\| < \delta$ , then  $\mathbf{x}_*$  is a local minimizer. Likewise, if there is a  $\delta > 0$  with*

$$\nabla f(\mathbf{x}_* + \mathbf{p})^\top \mathbf{p} < 0$$

*whenever  $\|\mathbf{p}\| < \delta$ , then  $\mathbf{x}_*$  is a local maximizer.*

# Multivariate Optimization Conditions

| Function Type | Local Minimizer                                                                          | Global Minimizer               | Reference                    |
|---------------|------------------------------------------------------------------------------------------|--------------------------------|------------------------------|
| $C^0$         | $f(x) \geq f(x_*)$ in some neighborhood                                                  | $f(x) \geq f(x_*)$ for all $x$ | Lec 2                        |
| $C^1$         | $\nabla f(x_*) = 0$ ,<br>$\nabla f(x)^\top (x - x_*) > 0$ in neighborhood                | Compare all local minimizers   | Theorem 1.1 ,<br>Theorem 3.1 |
| Convex        | $\nabla f(x_*) = 0$                                                                      | Any local minimizer is global  | Theorem 1.2                  |
| $C^2$         | $\nabla f(x_*) = 0$ and<br>$\nabla^2 f(x_*) \succeq 0$<br>(necessary but not sufficient) | ??                             | Theorem 2.1                  |

## Proof of the first order sufficient condition

We will just prove the local minimizer statement.

We may assume that  $f$  is differentiable in the  $\delta$ -neighborhood of  $\mathbf{x}_*$ .

Let  $\mathbf{0} \neq \mathbf{p} \in \mathbb{R}^n$  with  $\|\mathbf{p}\| < \delta$ .

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We will just prove the local minimizer statement.

We may assume that  $f$  is differentiable in the  $\delta$ -neighborhood of  $\mathbf{x}_*$ .

Let  $\mathbf{0} \neq \mathbf{p} \in \mathbb{R}^n$  with  $\|\mathbf{p}\| < \delta$ .

Define the univariate function  $g(t) = f(\mathbf{x}_* + t\mathbf{p})$ . Then  $g$  is continuous on  $[0, 1]$  and differentiable on  $(0, 1)$  with

$$g'(t) = \nabla f(\mathbf{x}_* + t\mathbf{p})^\top \mathbf{p}$$

## Proof of the first order sufficient condition

We will just prove the local minimizer statement.

We may assume that  $f$  is differentiable in the  $\delta$ -neighborhood of  $\mathbf{x}_*$ .

Let  $\mathbf{0} \neq \mathbf{p} \in \mathbb{R}^n$  with  $\|\mathbf{p}\| < \delta$ .

Define the univariate function  $g(t) = f(\mathbf{x}_* + t\mathbf{p})$ . Then  $g$  is continuous on  $[0, 1]$  and differentiable on  $(0, 1)$  with

$$g'(t) = \nabla f(\mathbf{x}_* + t\mathbf{p})^\top \mathbf{p}$$

By the Mean Value Theorem, there exists  $\xi \in (0, 1)$  such that  $g(1) - g(0) = g'(\xi)$ , i.e.,

$$\begin{aligned} f(\mathbf{x}_* + \mathbf{p}) - f(\mathbf{x}_*) &= \nabla f(\mathbf{x}_* + \xi\mathbf{p})^\top \mathbf{p} \\ &= \frac{1}{\xi} \nabla f(\mathbf{x}_* + \xi\mathbf{p})^\top (\xi\mathbf{p}) \\ &= \frac{1}{\xi} \nabla f(\mathbf{x}_* + \mathbf{q})^\top \mathbf{q} \end{aligned}$$

where we have written  $\mathbf{q} := \xi\mathbf{p}$ .



Since  $0 < \|\mathbf{q}\| = \xi\|\mathbf{p}\| < \|\mathbf{p}\| < \delta$ , we have  $\nabla f(\mathbf{x}_* + \mathbf{q})^\top \mathbf{q} > 0$ .

Hence  $f(\mathbf{x}_* + \mathbf{p}) > f(\mathbf{x}_*)$ .

Since this holds for all  $\mathbf{p}$  with  $\|\mathbf{p}\| < \delta$ , we conclude that  $\mathbf{x}_*$  is a local minimizer.

Note an equivalent way to state the condition " $\nabla f(\mathbf{x}_* + \mathbf{p})^\top \mathbf{p} > 0$  whenever  $\|\mathbf{p}\| < \delta$ " is:

$$\nabla f(\mathbf{x})^\top (\mathbf{x} - \mathbf{x}_*) > 0$$

whenever  $\|\mathbf{x} - \mathbf{x}_*\| < \delta$ . We have simply rewritten  $\mathbf{x} = \mathbf{x}_* + \mathbf{p}$ .

From the previous section, a necessary conditions for  $\mathbf{x}_*$  to be a local minimizer of a  $C^2$  scalar function are

$$\nabla f(\mathbf{x}_*) = \mathbf{0} \text{ and } \nabla^2 f(\mathbf{x}_*) \succeq 0.$$

Today we will show that sufficient conditions for  $\mathbf{x}_*$  to be a local minimizer are

$$\nabla f(\mathbf{x}_*) = \mathbf{0} \text{ and } \nabla^2 f(\mathbf{x}_*) \succ 0.$$

## Lemma

*Let  $\Omega$  be an open subset of  $\mathbb{R}^n$  or  $\mathbb{R}^{m \times n}$ , and let  $F : \Omega \rightarrow \mathbb{S}^n$  be a continuous function. If  $F(\mathbf{x}_*) \succ 0$  then  $F(\mathbf{x}) \succ 0$  for all  $\mathbf{x}$  in some neighborhood of  $\mathbf{x}_*$ , i.e., there exists  $\delta > 0$  such that  $F(\mathbf{x}) \succ 0$  whenever  $\|\mathbf{x} - \mathbf{x}_*\| < \delta$ .*

*This is the multivariate version of the fact that if a continuous function is positive at some point, then it is positive in some neighborhood of the point.*

### **Theorem (Second Order Sufficient Conditions)**

*Let  $f \in C^2(\Omega)$ . If  $\nabla f(\mathbf{x}_*) = \mathbf{0}$  and  $\nabla^2 f(\mathbf{x}_*) \succ 0$ , then  $\mathbf{x}_* \in \Omega$  is strict local minimizer.*

# Multivariate Optimization Conditions

| Function Type | Local Minimizer                                                           | Global Minimizer                                                                                  | Reference                    |
|---------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------|
| $C^0$         | $f(x) \geq f(x_*)$ in some neighborhood                                   | $f(x) \geq f(x_*)$ for all $x$                                                                    | Lec 2                        |
| $C^1$         | $\nabla f(x_*) = 0$ ,<br>$\nabla f(x)^\top (x - x_*) > 0$ in neighborhood | Compare all local minimizers                                                                      | Theorem 1.1 ,<br>Theorem 3.1 |
| Convex        | $\nabla f(x_*) = 0$                                                       | Any local minimizer is global                                                                     | Theorem 1.2                  |
| $C^2$         | $\nabla f(x_*) = 0$ and<br>$\nabla^2 f(x_*) \succ 0$                      | $\nabla f(x_*) = 0$ ,<br>$\nabla^2 f(x_*) \succ 0$ , and<br>$\nabla^2 f(x) \succeq 0$ for all $x$ | Theorem 2.1,<br>Theorem 3.2  |

## Proof of the second order sufficient condition

Choose  $\varepsilon > 0$  so that  $\mathbf{x}_* + \mathbf{p} \in \Omega$  for all  $\|\mathbf{p}\| < \varepsilon$ . Then for any such  $\mathbf{p} \neq \mathbf{0}$ ,

$$f(\mathbf{x}_* + \mathbf{p}) = f(\mathbf{x}_*) + \nabla f(\mathbf{x}_*)^\top \mathbf{p} + \frac{1}{2} \mathbf{p}^\top \nabla^2 f(\mathbf{x}_* + \xi \mathbf{p}) \mathbf{p}$$

for some  $\xi \in (0, 1)$ .

Since  $\nabla^2 f : \Omega \rightarrow \mathbb{S}^n$  is continuous, Lemma 3.1 implies that  $\nabla^2 f(\mathbf{x}) \succ 0$  in a neighborhood of  $\mathbf{x}_*$ .

If we pick  $\varepsilon$  small enough, then  $\nabla^2 f(\mathbf{x}_* + \xi \mathbf{p}) \succ 0$ . Hence

$$f(\mathbf{x}_* + \mathbf{p}) = f(\mathbf{x}_*) + \underbrace{\nabla f(\mathbf{x}_*)^\top \mathbf{p}}_{=0} + \underbrace{\frac{1}{2} \mathbf{p}^\top \nabla^2 f(\mathbf{x}_* + \xi \mathbf{p}) \mathbf{p}}_{>0}$$

Therefore

$$f(\mathbf{x}_* + \mathbf{p}) > f(\mathbf{x}_*)$$

for all  $\mathbf{p}$  in some neighborhood of  $\mathbf{x}_*$ . Hence  $\mathbf{x}_*$  is a strict local minimizer.  $\therefore$





| Function Type | Local Minimizer                                                           | Global Minimizer                                                                                  | Reference                    |
|---------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------|
| $C^0$         | $f(x) \geq f(x_*)$ in some neighborhood                                   | $f(x) \geq f(x_*)$ for all $x$                                                                    | Lec 2                        |
| $C^1$         | $\nabla f(x_*) = 0$ ,<br>$\nabla f(x)^\top (x - x_*) > 0$ in neighborhood | Compare all local minimizers                                                                      | Theorem 1.1 ,<br>Theorem 3.1 |
| Convex        | $\nabla f(x_*) = 0$                                                       | Any local minimizer is global                                                                     | Theorem 1.2                  |
| $C^2$         | $\nabla f(x_*) = 0$ and<br>$\nabla^2 f(x_*) \succ 0$                      | $\nabla f(x_*) = 0$ ,<br>$\nabla^2 f(x_*) \succ 0$ , and<br>$\nabla^2 f(x) \succeq 0$ for all $x$ | Theorem 2.1,<br>Theorem 3.2  |